ch27 Nuclear Energy

2022 Q33

33. What is the value of x for the fission reaction below ?

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{92}_{36}\mathrm{Kr} + ^{141}_{56}\mathrm{Ba} + x\,^{1}_{0}\mathrm{n}$$

A. 1
B. 2
C. 3
D. 4

2018 Q31

31. Which of the following nuclear reactions is/are possible for a chain reaction to occur ?

(1) $^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{1}_{0}\mathrm{n} + ^{4}_{2}\mathrm{He}$

(2) $^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{141}_{56}\mathrm{Ba} + ^{92}_{36}\mathrm{Kr} + 3\,^{1}_{0}\mathrm{n}$

(3) $^{239}_{94}\mathrm{Pu} + ^{1}_{0}\mathrm{n} \rightarrow ^{148}_{58}\mathrm{Ce} + ^{89}_{36}\mathrm{Kr} + 3\,^{1}_{0}\mathrm{n}$

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2025 Q33

*33. In the following nuclear reaction, nuclide X and Y fuse to form nuclide Z under suitable conditions and the amount of energy released is 7.16 MeV.

$$X + Y \rightarrow Z$$

The total mass of X and Y is

A. 0.00385 u less than the mass of Z.
B. 0.00385 u more than the mass of Z.
C. 0.00769 u less than the mass of Z.
D. 0.00769 u more than the mass of Z.

2017 Q33

*33. The following shows the decay of uranium-238 ($^{238}_{92}U$).

$$^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4}_{2}He$$

Given that : mass of $^{238}_{92}U$ = 238.05079 u

mass of $^{234}_{90}Th$ = 234.04363 u

mass of $^{4}_{2}He$ = 4.00260 u

Which of the following statements is/are correct ?

(1) The temperature required to start the decay is about 10^7 K.
(2) The energy released in the decay of one uranium-238 nucleus is 4.25 MeV.
(3) All the energy released in the decay becomes the kinetic energy of $^{4}_{2}He$.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2019 Q33

*33. Given: mass of a neutron = 16749×10^{-31} kg
mass of a proton = 16726×10^{-31} kg
mass of an electron = 9×10^{-31} kg

In a nuclear reaction, a neutron becomes a proton and a β particle. Estimate the energy released in this process.

A. 1.8 MeV
B. 1.3 MeV
C. 0.79 MeV
D. 0.51 MeV

2013 Q36

*36. The sun releases huge amount of energy through thermonuclear fusion while at the same time its mass decreases. The average power released by the sun is about 3.8×10^{26} W. Estimate the decrease in mass of the sun in one second.

A. 4.2×10^{6} kg
B. 4.2×10^{9} kg
C. 1.3×10^{15} kg
D. 1.3×10^{18} kg

2018 Q33

*33. Given: mass of proton = 1.007276 u
mass of neutron = 1.008665 u
mass of ${}^{3}_{2}He$ nucleus = 3.016030 u
1 u = 931 MeV

When a ${}^{3}_{2}He$ nucleus is formed from 2 protons and 1 neutron,

A. 6.7 MeV energy is released.
B. 6.7 MeV energy is required.
C. 8.0 MeV energy is released.
D. 8.0 MeV energy is required.

2014 Q33

*33. A radium nucleus decays to a radon nucleus by emitting an α-particle. The energy released in the process is 4.9 MeV. Compared to the mass of a radium nucleus, the total mass of a radon nucleus and an α-particle is

A. 5.4×10^{-11} kg less.
B. 5.4×10^{-11} kg more.
C. 8.7×10^{-30} kg less.
D. 8.7×10^{-30} kg more.

2023 Q32

*32. When a proton (^1_1H) bombards a lithium nucleus (^7_3Li), two α particles (^4_2He) are produced.

$$^1_1H + ^7_3Li \rightarrow 2\ ^4_2He$$

The energy released in this nuclear reaction is 17.32 MeV. Find the mass of a lithium nucleus.

Given: mass of ^1_1H = 1.0078 u

mass of ^4_2He = 4.0026 u

A. 6.9788 u
B. 6.9974 u
C. 7.0017 u
D. 7.0160 u

2023 Q33

33. Which statement about nuclear fission is correct ?

A. It involves two nuclei combined together.
B. All nuclear fission reactions are spontaneous.
C. Rate of fission reaction is temperature dependent.
D. Fission of heavy nuclei yields products that are more stable in terms of energy.

2026 Q33

*33. Consider the following nuclear reaction:

$$^2_1H + ^3_1H \rightarrow ^4_2He + ^1_0n$$

Using the data below, estimate the energy released when 5 g of ^2_1H completely reacts with ^3_1H through the reaction.

mass of ^2_1H = 2.0136 u

mass of ^3_1H = 3.0155 u

mass of ^4_2He = 4.0015 u

mass of ^1_0n = 1.0087 u

A. 3×10^{-12} J
B. 3×10^{-10} J
C. 4×10^{12} J
D. 2×10^{18} J

pp Q36

*36. For the following nuclear reaction, state the type of reaction and determine the energy released.

$${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$$

Given: mass of ${}^{2}_{1}H = 2.014$ u

mass of ${}^{3}_{1}H = 3.016$ u

mass of ${}^{4}_{2}He = 4.003$ u

mass of ${}^{1}_{0}n = 1.009$ u

	Type of reaction	Energy released
A.	fusion	0.018 MeV
B.	fusion	16.76 MeV
C.	fission	0.018 MeV
D.	fission	16.76 MeV

sap Q34

34. Which of these is a nuclear fusion reaction ?

A. ${}^{235}_{92}U + n \rightarrow {}^{144}_{56}Ba + {}^{90}_{36}Kr + 2n$

B. ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n$

C. ${}^{14}_{7}N + n \rightarrow {}^{14}_{6}C + {}^{1}_{1}H$

D. ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + \alpha$